

Legend

Formalized
Result

Formalized
Lemma

Conditional
Bridge

Definition

Open /
Pending

Model Definitions

Utilities, policies, fairness

Lemma 1 (App. C)

$I_{\min}^* > 0$

Lemma 2 (App. C)

Item-fairness LP epigraph

Proposition 2

Symmetric optimum exists

Reduction Bridges

Original $\leftrightarrow$ reduced model

Theorem 3

First-half price monotonicity

Problem 6

Opposing-type
LP translation

Lemma 4 (App. D)

Basic support +
no-gap structure

Lemma 5 (App. D)

Closed-form policy family

Lemmas 6–11 (App. D)

Pivot, mono-
tonicity, stitching

Lemma 16 (App. E)

q_j / midpoint algebra

Problem 11 (App. E)

Three-type LP, $\gamma = 1$
bridge, real-BFS interface

Lemma 12 (App. E)

Symmetric reduction into S'

Lemma 17 (App. E)

No right-half type-zero mass

Lemma 13 Perturbations

x -gap and z -gap certificates

Lemma 13 (App. E)

Pivot support from
equalized optimum

Lemma 15 (App. E)

Coordinates; half-
LP center λ caveat

Lemma 14 (App. E)

Equality-form op-
timum uniqueness

Theorem 4 No-Extremes

Equality-form Prob-
lem 11 bridge

Cold-Start Bounds

No-extreme $\Rightarrow$ low utility

No-Fairness Side

Estimated $\gamma = 0$ benchmark

True-Model Lower Bound

$U_{\min}^*(1, w) > 1/n$

Theorem 4 Wrappers

Formalized from equality-
form Problem 11

Value-Vector Witness

Geometric $v_j = r^j$
construction

Theorem 4 (App. E)

Open: real equality-
form BFS existence